

## EX. NO. 10 - GHIDRA EXECUTABLE ANALYSIS

|                 |                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Experiment Name | Use Ghidra to Disassemble and Analyze an Executable                                                                                                                                               |
| Aim             | To use Ghidra to disassemble and analyze an executable file, examine its functions, strings, imported components, and decompiled code, and understand the basic behavior of the analyzed program. |
| Tool Used       | Ghidra 12.1.4, Java Development Kit (JDK), Windows operating system, Benign test executable (sample.exe)                                                                                          |

### OBJECTIVE

- To understand the basic use of Ghidra for executable analysis.
- To examine the structure and components of a Portable Executable (PE) file.
- To identify strings, imported components, and functions using Ghidra.
- To examine the main function using the CodeBrowser and Decompiler.
- To understand the basic workflow used in static malware analysis.

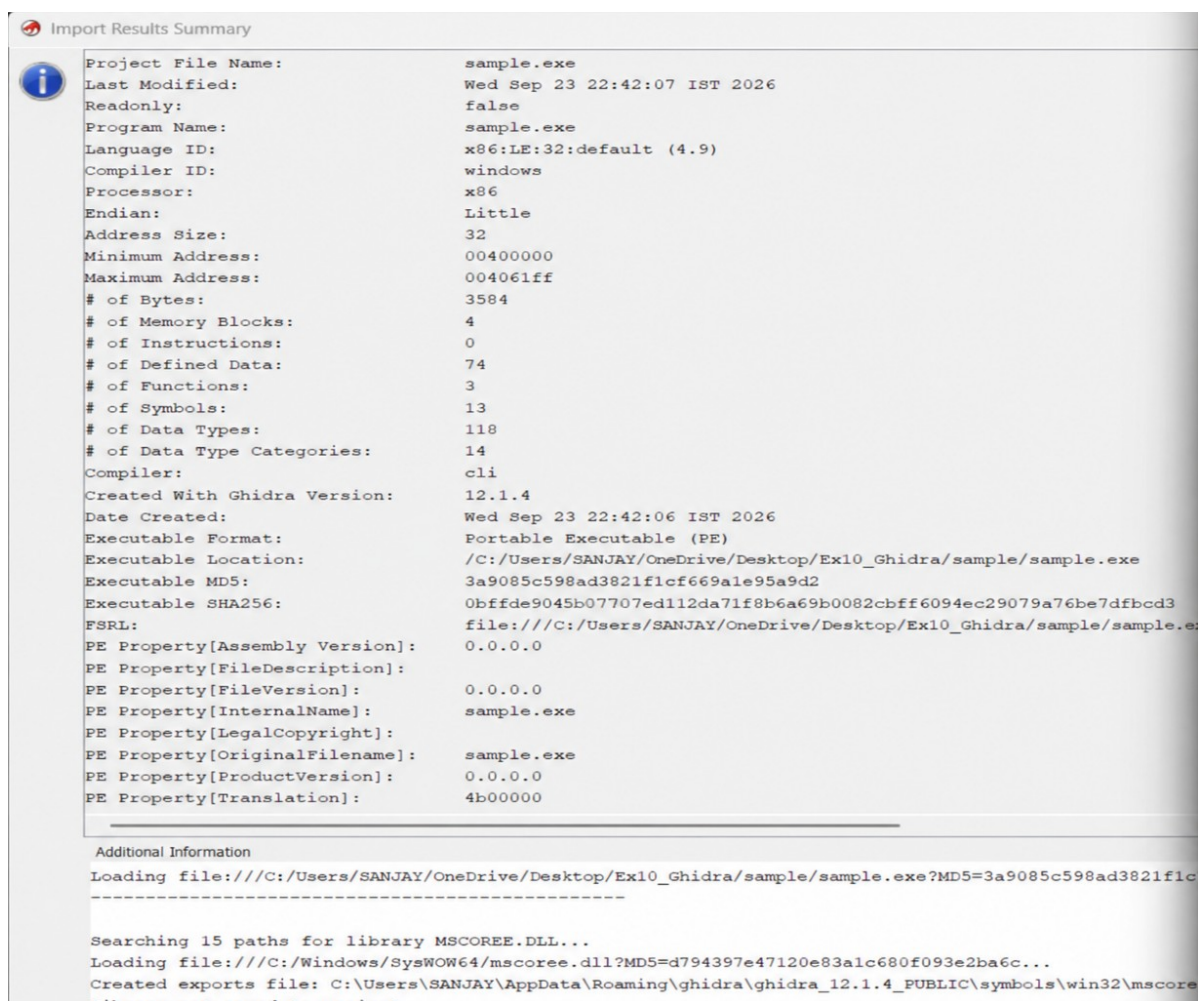
### SAFETY NOTE

A harmless test executable was used for this experiment instead of live malware. The executable was analyzed statically without intentionally executing it.

### PROCEDURE

#### Step 1 - Create a Ghidra Project

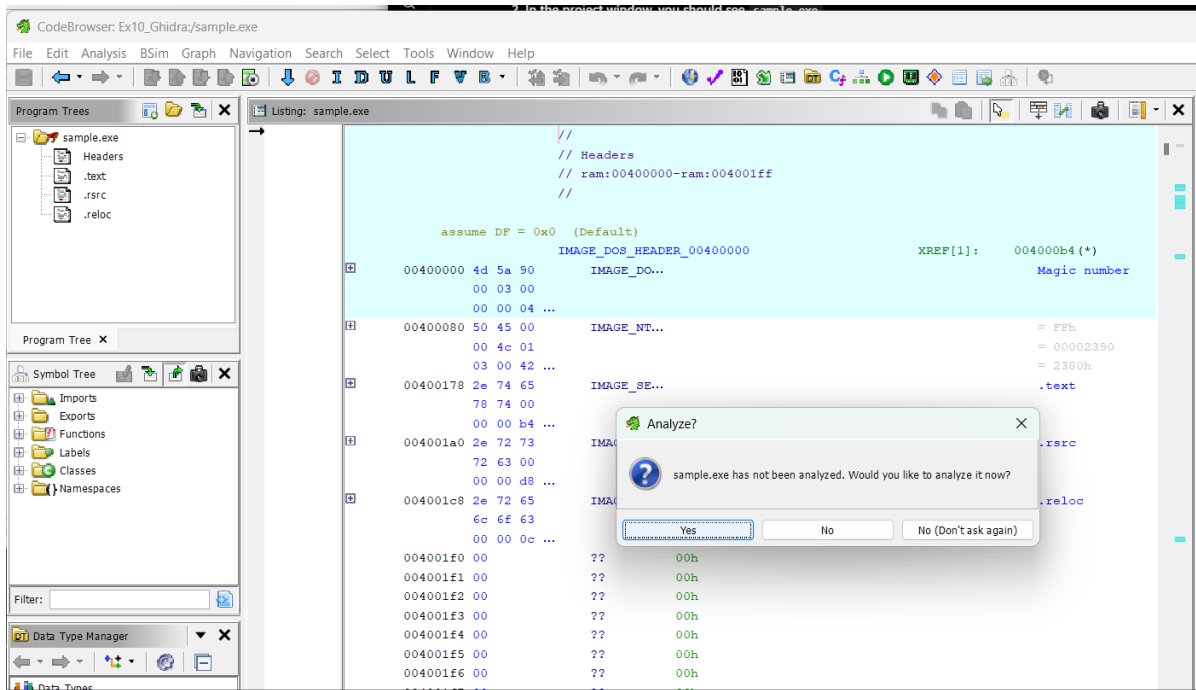
A new non-shared Ghidra project named Ex10\_Ghidra was created.



Screenshot 1 - Ghidra project/file information

#### Step 2 - Import the Executable

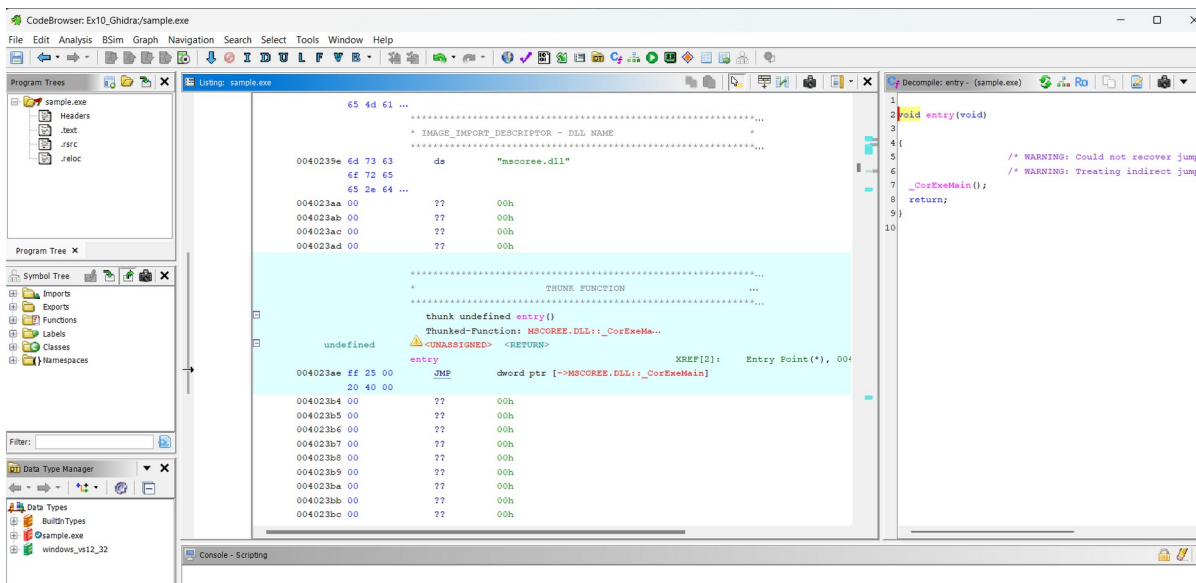
The sample.exe file was imported into the Ghidra project. Ghidra identified the file as a Portable Executable (PE).



Screenshot 2 - Executable imported into Ghidra

### Step 3 - Perform Automatic Analysis

The default Ghidra analysis options were used to analyze the executable.



Screenshot 3 - Automatic analysis

### Step 4 - Analyze Strings

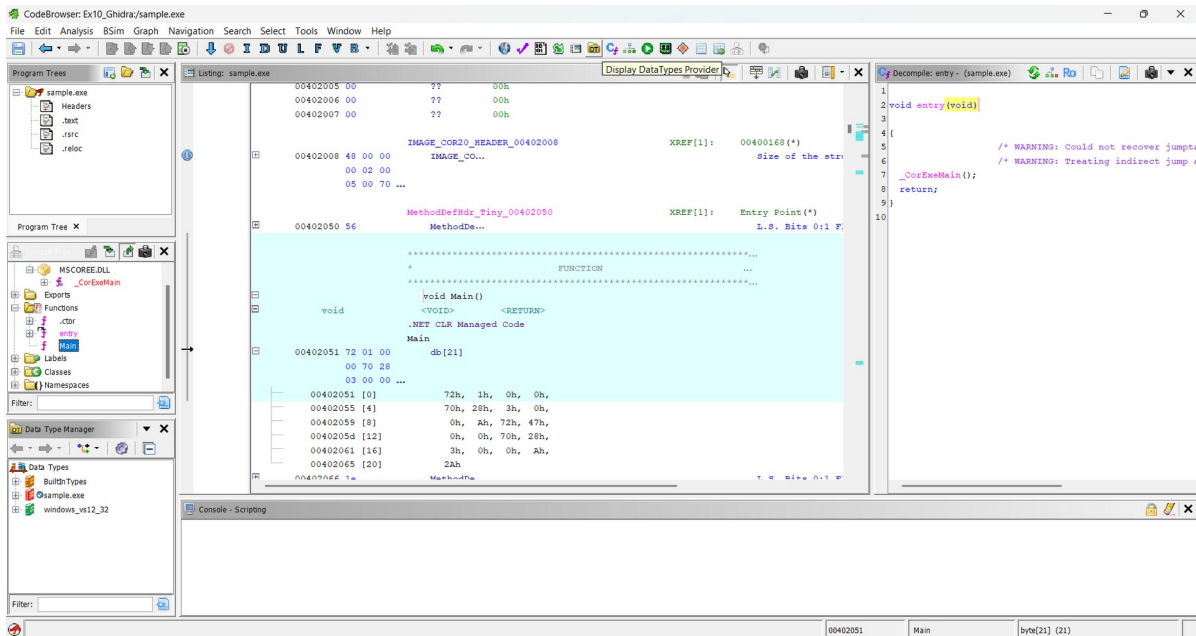
The String Search feature was used to identify readable strings in the executable. Ghidra identified 44 strings.

Important strings included:

- Ghidra Malware Analysis Lab Sample
- This is a harmless test program.
- Console
- WriteLine

### Step 5 - Analyze Imported Components

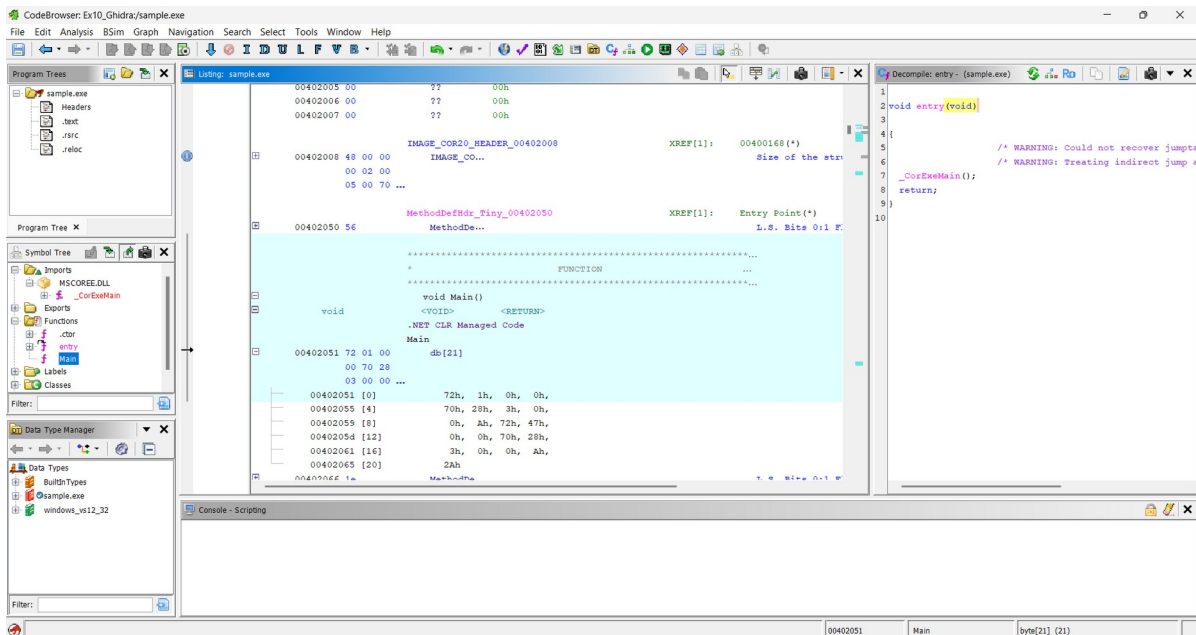
The Imports section was examined to identify external components used by the executable. The analysis showed MSCORE...DLL and \_CorExeMain.



**Screenshot 4 - Imported components**

## Step 6 - Analyze Functions

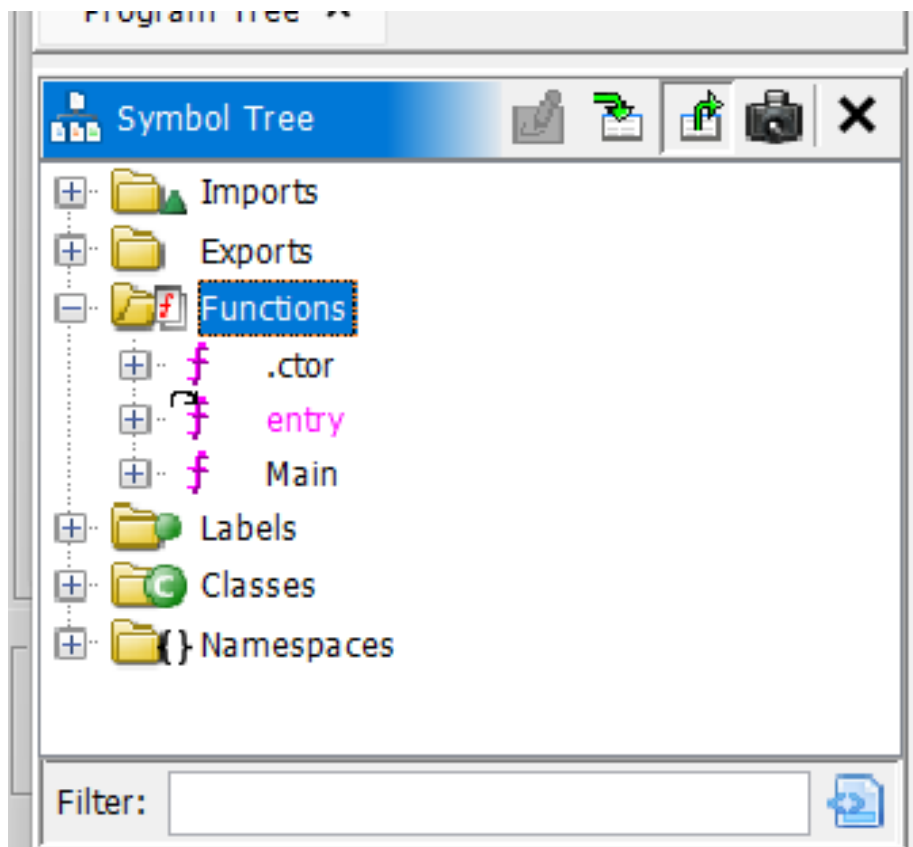
The Functions section was examined. Ghidra identified the functions .ctor, entry, and Main.



**Screenshot 5 - Functions identified by Ghidra**

## Step 7 - Analyze the Main Function

The Main function was opened in the Ghidra CodeBrowser and examined using the available decompiler view.



Screenshot 6 - Main function / Symbol Tree

## OBSERVATION

The executable was successfully loaded and analyzed using Ghidra. The file was identified as a Portable Executable (PE) for the x86 architecture. Ghidra identified the .text, .rsrc, and .reloc sections and detected 44 readable strings. The imported runtime component included MSCORE...DLL with \_CorExeMain. The Functions window identified .ctor, entry, and Main.

The strings found in the executable indicated that it was a harmless test program created for analysis purposes.

## RESULT

The sample.exe file was successfully imported, disassembled, and analyzed using Ghidra. The executable's strings, imported components, functions, and Main function were examined using the Ghidra CodeBrowser and available decompiler views.